Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad Cantonment, Natore-6431

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Sessional Plan

Program:	Computer Science and Engineering (CSE)
Course Code & Title:	**CSE-2106 & Database Systems Laboratory**
Semester:	Summer-2025
Level:	2nd Year 1st Semester
Credit Hour:	3.00 Hr.
Curriculum Structure	Outcome Based Education (OBE)
Name & Designation of Teacher(s):	**Md. Omar Faruq** Assistant Professor, Dept. of CSE **Md. Rafiqul Islam (Rafiq)** Lecturer, Dept. of CSE
Class Hours:	Monday: 11:40 AM -02:55 PM [Group 1] & Tuesday: 11:40 AM -02:55 PM [Group 2]
Consultation Hours:	Wednesday: 11:40 AM -02:55 PM Monday: 08:15 AM -11:40 AM
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Rationale:	This course is designed to introduce the basic concepts of database, learn the foundations of database systems, focusing on basics such as the relational algebra and data model, schema normalization, query optimization, and database designing.
Pre-requisite (if any):	N/A

Objectives:

1. Understand the basic concepts and appreciate the applications of database systems.
2. Know the basics of SQL and construct queries using SQL.
3. Be familiar with a commercial relational database system (MySQL/Oracle) by writing SQL using the system.
4. Be familiar with the relational database theory, and be able to write relational algebra expressions for queries.
5. To learn the design of a database starting from the conceptual design to the implementation of database schemas and user interfaces to a database.

Course Content:

Introduction to SQL: Overview of the SQL query, Language, SQL data definition, Basic structure of SQL, Queries, Null values, Aggregate functions, Modification of the database.

Intermediate SQL: Join expressions, Views, Transactions, Integrity constraints, SQL data types and schemas.

Advanced SQL: Accessing SQL from a programming language, Functions and procedures, Triggers, Recursive Queries, Advanced aggregation features.

Text Book(s):

1. DatabaseSystemConcepts,Mcgraw-Hill-A.Silberschatz,7th Edition,ISBN- 9780078022159
2. DatabaseManagementSystem,McGraw-HillHigherEducation-R.Ramakrishnan,
3. JohannesGehrke,3rd Edition,ISBN-0-07-24-6563-8

Reference Book(s):

1. Principles of Database Management, Prentice-hall-James Martin
2. Database Management systems, Prentice-Hall Publication-Ullman
3. Oracle8iaBeginnersGuide,McGrawHill-Abey
4. SQL, PL/SQL – The Programming Language of Oracle, Ivan Bayross

*** Additional reading material: PowerPoint Slides & PDFs provided by the teachers**

Student's responsibilities:

a. Students must come to the class prepared for the course material covered in the previous class.
b. They must submit their Lab report, lab project, Lab task / assignments on time.
c. No late or partial Lab report, Lab task / assignments will be acceptable.

Minimum Attendance: 80% class attendance is mandatory for a student in order to appear at the final examination.

Teaching-learning and Assessment Strategy:
Sessional class participations (Class Performance/ Continuous Evaluation), Lab Reports, Lab Project/Lab Test, Quiz and Viva-voce

Grading System: As per the approved grading scale of Bangladesh Army University of Engineering & Technology (BAUET), Qadirabad, Natore **(Appendix-II)**

Course Assessment Methods & their Weights:

	Marks (%)
Conduct of Lab test/Lab Performance	25%
Report Writing/ Programming	15%
Mid Term Evaluation (Exam/Project/Assignment)	20%
Final Evaluation (Exam/Project/Assignment)	30%
Viva Voce/Presentation Skill	10%
Total	**100%**

Grading Policy:

Numerical Score	Letter Grade		Grade points
80% and above	A+	(APlus)	4.00
75% to less than 80%	A	(ARegular)	3.75
70% to less than 75%	A-	(AMinus)	3.50
65% to less than 70%	B+	(BPlus)	3.25
60% to less than 65%	B	(BRegular)	3.00
55% to less than 60%	B-	(BMinus)	2.75
50% to less than 55%	C+	(CPlus)	2.50
45% to less than 50%	C	(CRegular)	2.25
40% to less than 45%	D	(DPass)	2.00
Less than 40%	F	Fail	0.00
Incomplete	I	-	-
Withdrawal	W	-	-
Project/Thesis Continuation	X	-	-

Detailed Mapping (According to BAETE Accreditation Manual- II)

i. Mapping CO-PO, Bloom's Level, Knowledge and Attitude Profile (WK), Complex Engineering Problem Solving (WP) and Complex Engineering Activities (EA)

COs	CO Statements Upon completion of the course, the students will be able to)	PO	WK	WP	EA	Bloom's Level	Assessment Tools
CO1	Demonstrate the knowledge in projects with a commercial relational database system (Oracle) and design a team-based project.	1,8	5 7	2	1	C3, P1	Lab test/Project, Written test/Quiz, Viva Voce, Attendance, Data Sheet/Home Assignments/ Report, Class Performance /Continuous Evaluation)
CO2	Illustrate the basics of SQL and construct queries using SQL.	3	3	1	1	C2, P2	
CO3	Be familiar with a commercial relational database system (Oracle/MySQL) by writing SQL using the system.	5	6	2	1	C2,C3	
CO4	Apply SQL integration with programming languages to implement different advance database functionalities	2,6	4	5	1	P5	

Legends: CO-Course Outcomes PO-Program Outcomes, KP-Knowledge Profile
Bloom's Level: Cognitive Domain (C1-C6), Psychomotor Domain (P1-P7), and Affective Domain (A1-A5)

COs	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1	M											
CO2					H							
CO3			H									
CO4		M										

Legends: High (H)=3, Medium(M)=2 and Low(L)=1

Detailed Mapping:

Course Learning Outcomes (COs)	CO1	CO2	CO3	CO4
Program Outcomes (PO)				
PO1 Engineering knowledge	X			
PO2 Problem Analysis				X
PO3 Design/ development of solutions			X	
PO4 Investigation				
PO5 Modern Tool Usage		X		
PO6 The Engineer and Society				
PO7 Environment and Sustainability				
PO8 Ethics				
PO9 Individual and Team work				
PO10 Communication				
PO11 Project Management and Finance				
PO12 Lifelong learning				
PO Marks Ratio	20%	30%	50%	
Blooms Taxonomy level/Domain level	C1	P4	C5	
Assessment tools	Lab Report, Quiz viva, Assignment, Presentation, Lab test			
Knowledge Profile (K)				
K1 Natural sciences				
K2 Mathematics				
K3 Engineering Fundamentals (PO1-PO2)	X			X
K4 Specialist knowledge (PO1-PO2)		X		
K5 Engineering Design (PO3)				
K6 Engineering practice (PO5)			X	
K7 Comprehension (PO6-PO8)				
K8 Research literature (PO4)				
Complex Engineering Problems (P)				
P1 Depth of Knowledge required (PO1- PO7)	X			
P2 Range of conflicting requirements (PO1- PO7)		X		
P3 Depth of analysis required (PO1- PO7)				
P4 Familiarity of issues (PO1- PO7)				
P5 Extent of applicable codes (PO1- PO7)			X	
P6 Extent of stakeholder (PO1- PO7)				
P7 Interdependence (PO1- PO7)				
Complex Engineering Activities (A)				
A1 Range of resources (PO10)	X	X	X	X
A2 Level of interactions (PO10)				
A3 Innovation (PO10)				
A4 Consequences to society and the environment (PO10)				
A5 Familiarity (PO10)				

PO Distribution:

COs	CO Statements Upon completion of the course, the students will be able to)	PO (100)
CO1	Demonstrate the knowledge in projects with a commercial relational database system (Oracle) and design a team-based project.	10%
CO2	Illustrate the basics of SQL and construct queries using SQL.	30%
CO3	Be familiar with a commercial relational database system (Oracle/MySQL) by writing SQL using the system.	50%
CO4	Apply SQL integration with programming languages to implement different advance database functionalities	10%

i. Assessment of CO-PO, Bloom's Level

Teaching-learning and Assessment Strategy:

Lab test/Project, Written test/Quiz, Viva Voce, Attendance, Data Sheet/Home Assignments/Report, Class Performance /Continuous Evaluation

Assessment Methods	Description	Grading	CO	Blooms Taxonomy
Continuous Assessment (100%)	Conduct of Lab test/Lab Performance	25%	CO2	C2, P2
	Report Writing/ Programming	15%	CO1, CO2,CO3, CO4	C2, C3, P1, P2, P5
	Mid Term Evaluation (Exam/Project/Assignment)	20%	CO2,CO3	C2, C3, P2
	Final Evaluation (Exam/Project/Assignment)	30%	CO2,CO3	C2, C3, P2
	Viva Voce/Presentation Skill	10%	CO2,CO3	C2, C3, P2
	Total	**100%**		

Generic Skills: Graduate Attributes/Program Outcomes (POs)

Program Outcomes (POs) represent the knowledge, skills and attitudes the students should have at the end of a four-year engineering program. CSE program of BAUET has 12 Program Outcomes. They are briefly described in the following table.

SL	POs	Category	Description
1	**PO 1**	**Engineering Knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2	**PO 2**	**Problem Analysis**	Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	**PO 3**	**Design/Development of Solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, social and environmental concerns.
4	**PO 4**	**Investigation**	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
5	**PO 5**	**Modern Tool Usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6	**PO 6**	**The Engineer and Society**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	**PO 7**	**Environment and Sustainability**	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of need for sustainable development.
8	**PO 8**	**Ethics**	Apply ethical principles and commit to professional Ethics and responsibilities and norms of the engineering practice.
9	**PO 9**	**Individual and Team Work**	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10	**PO 10**	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**PO 11**	**Project Management and Finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12	**PO 12**	**Life Long Learning**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Pos	Graduate Attributes (Keywords)	A-K's	P/A
1	Engineering Knowledge	K1-K4	P
2	Problem Analysis	K1-K4	P
3	Design/Development of Solutions	K5	P
4	Investigation	K8	P
5	Modern Tool Usage	K6	P
6	The Engineer and Society	K7	P
7	Environment and Sustainability	K7	P
8	Ethics	K7	--
9	Individual and Team work	--	--
10	Communication		EA
11	Project Management and Finance	---	--
12	Life Long Learning	--	--
Assessments provide adequate feedback to the program to identify strengths and weaknesses for CQI			

Complex Engineering Problems (Need High Taxonomy Level)

Complex Engineering Problems have characteristic P1 and some or all of P2 to P7, EP1 and EP2, that can be resolved with in-depth forefront knowledge

P1	Depth of Knowledge required	Resolved with **forefront in-depth** engineering knowledge (K3, K4, K5, K6 or K8) which allows a fundamentals-based, first principles analytical approach
P2	Range of conflicting requirements	Involve **wide-ranging or conflicting** technical, engineering and other issues.
P3	Depth of analysis required	Have **no obvious solution** and require abstract thinking, originality in analysis to formulate suitable models.
P4	Familiarity of issues	Involve **infrequently encountered** issues
P5	Extent of applicable codes	**Beyond codes** of practice
P6	Extent of stakeholder involvement and level of conflicting requirements	Involve **diverse groups of stakeholders** with widely **varying needs**.
P7	Interdependence	Are **high level problems** including **many component** parts or sub-problems.

EP1	Consequences	Have **significant consequences** in a range of contexts.
EP2	Judgment	Require judgment in **decision** making

Complex Engineering Activities (Project based)

Activities	Preamble	**Complex activities** mean (engineering) activities or projects that have **some or all** of the following characteristics listed below
A1	Range of resources	**Diverse resources** (people, money, equipment, materials, information and technologies).
A2	Level of interaction	Require resolution of significant problems arising from interactions between **wide ranging** or **conflicting** technical, engineering or other issues.
A3	Innovation	Involve creative use of engineering principles and **research-based** knowledge in **novel** ways.
A4	Consequences to society and the environment	Have **significant consequences** in a **range of contexts**, characterized by **difficulty** of prediction and mitigation.
A5	Familiarity	Can extend **beyond previous** experiences by applying **principles-based** approaches.

Bloom's Taxonomy Domain:

Cognitive Domain		**Psychomotor Domain**		**Affective Domain**	
C1	Remembering	**P1**	Perception	**A1**	Receive
C2	Understanding	**P2**	Set	**A2**	Respond
C3	Applying	**P3**	Guided Response	**A3**	Value
C4	Analyzing	**P4**	Mechanism	**A4**	Organize
C5	Evaluating	**P5**	Complex Overt Response	**A5**	Internalize
C6	Creating/Designing	**P6**	Adaption		
		P7	Origination		

Sessional Lecture Plan

Sl. No.	Content of Course	Lecture Hours	COs	Teaching Aids	Teaching Assessment Tools
01	Implementation of DDL commands of SQL with suitable examples. ▪ Create table ▪ Alter table ▪ Drop table ▪ Rename table	03 Hrs.	CO1 CO2 CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation
02	Implementation of DML commands of SQL with suitable examples. ▪ Insert ▪ Update ▪ Delete	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion, Class Performance/ Continuous Evaluation
03	Implementation of different types of functions with suitable examples ▪ Number function ▪ Aggregate Function ▪ Character Function/ String Matching ▪ Conversion Function ▪ Date Function	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion, Class Performance/ Continuous Evaluation
04	Implementation of different types of operators in SQL. ▪ Arithmetic Operators ▪ Logical Operators ▪ Comparison Operator ▪ Special Operator ▪ Set Operation	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion, Class Performance/ Continuous Evaluation
05	Implementation of different types of Joins. ▪ Left Outer Join ▪ Right Outer Join ▪ Inner Join ▪ Full outer Join	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation

06	Study & Implementation of ▪ Group by & Having Clause ▪ Order by Clause ▪ Indexing.	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Lab Report, Individual or Group Discussion, Class Performance/ Continuous Evaluation
07	Mid Term Evaluation	03 Hrs.	CO2, CO3		
08	Study & Implementation of ▪ Sub queries ▪ Views	03 Hrs.	CO2 CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation
09	Study & Implementation of different types of constraints. ▪ Not null, Unique, Check ▪ Primary Key, Foreign Key ▪ Default	03 Hrs.	CO2 CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation
10	Study & Implementation of ▪ Database Backup & Recovery commands. ▪ Rollback, Commit, Savepoint.	03 Hrs.	CO2 CO3	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation
11	▪ Creating Database /Table Space ▪ Managing Users: Create User, Delete User ▪ Managing roles:-Grant, Revoke.	03 Hrs.	CO4	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation
12	Study & Implementation of SQL Triggers.	03 Hrs.	CO2	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation
13	Access and manipulation of database from web pages.	03 Hrs.	CO4	Power Point slides, Lab Manual, White Board	Sessional Class, Individual or Group Discussion, Class Performance/ Continuous Evaluation
14	Final Evaluation	03 Hrs.	CO2, CO3		

Instructions for the students to conduct Sessional Class

SL	Instructions
1.	Experiment list for the sessional has been displayed. Each experiment is given a number.
2.	For each day, every student must have to implement the given experimental task according to experiment number. Marks will be given by the course teacher for the implemented experiment.
3.	Students may be permitted to carry out missed experiments in their own time with the approval of course teacher only.
4.	Sessional report must be submitted on the lab final. Following Criteria will be included in the report book. a. Experiment No b. Experiment name c. Objectives d. Theory e. Source Code f. Input g. Output h. Conclusion/Discussion
5.	Sessional marks awarded at the end of the semester will be strictly followed according to the sessional assessment method.

Sample Questions for Sessional Task

game

id	mdate	stadium	team1	team2
1001	8 June 2012	National Stadium, Warsaw	POL	GRE
1002	8 June 2012	Stadion Miejski (Wroclaw)	RUS	CZE
1003	12 June 2012	Stadion Miejski (Wroclaw)	GRE	CZE
1004	12 June 2012	National Stadium, Warsaw	POL	RUS
...				

goal

matchid	teamid	player	gtime
1001	POL	Robert Lewandowski	17
1001	GRE	Dimitris Salpingidis	51
1002	RUS	Alan Dzagoev	15
1002	RUS	Roman Pavlyuchenko	82
...			

world

name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000000
Albania	Europe	28748	2831741	12960000000
Algeria	Africa	2381741	37100000	188681000000
Andorra	Europe	468	78115	3712000000
Angola	Africa	1246700	20609294	100990000000
...				

Construct the following query using the above table:

i. Show matchid and player name for all goals scored by Poland.
ii. Show player, teamid, coach, gtime for all goals scored in the first 15 minutes (gtime=10)
iii. Show the player, teamid and mdate and for every Italian goal.
iv. Show the team1, team2 and player for every goal scored by a player called Messi.
v. Show the name and the per capita GDP for those countries with a population of at least 150 million.
vi. Show the name and population in millions for the countries of the continent 'Africa'. (Hints: Divide the population by 1000000 to get population in millions.)
vii. For each continent show the continent and number of countries.
viii. List the continents that have a total population of at least 200 million.
ix. Show the number of countries with population greater than 400000.
x. Show the average population of ‘Bangladesh’, ‘India’ and 'Pakistan'.

Assignment

Full Marks: 10

Question	CO	PO	BL	P	Marks
	1,4	1,2	C3,P5	3	10

Step-1: Define the problem

Range of Conflicting Requirements (P2)

Database systems are designed to manage large bodies of information. These large bodies of information do not exist in isolation. Design a database management system (choose unique database name) with entities such as teachers, students, courses and etc. for any relationships. Such as each student associated with a teacher for advisor and students take courses that conducted by the teachers.

- The entities consist of the primary key attribute, complex attribute, and composite attribute, multi-valued attribute, stored and derived attribute.
- Add entity and project then formulate the relational schema from the database design.
- Construct the Entity-Relationship (E-R) diagram and database schema of your unique database system.

Step-2: Test and Implement a Solution

Extend of Applicable Codes (P5)

- Create the database using the data definition language
- Insert the values and represent your unique database using data manipulation language.
- Make use of the manipulation language to prove your database is easy to use, store, and retrieve information that is both convenient and efficient. [CO1,4-C3,P5]

Approved by

Head

Dept. of CSE, BAUET